

Calculus II

Complex multiplication

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Let $u = 2 + 3i$, $v = 5 - 7i$.

Example (Addition)

$$u + v = (2 + 3i) + (5 - 7i) = (2 + 5) + (3 - 7)i = 7 - 4i.$$

Example (Subtraction)

$$u - v = (2 + 3i) - (5 - 7i) = (2 - 5) + (3 - (-7))i = -3 + 10i.$$

Example (Multiplication)

$$\begin{aligned}u \cdot v &= (2 + 3i) \cdot (5 - 7i) \\&= 2 \cdot 5 + 2 \cdot (-7)i + 3i \cdot 5 + 3i(-7i) \\&= 10 - 14i + 15i - 21i^2 \\&= 10 + i - (-21) \\&= 31 + i\end{aligned}$$